

- Q.20 What is meant by a problem space? Explain. (CO1)
- Q.21 Highlight the use of mappings in AI. (CO3)
- Q.22 Describe the process of resolution in predicate logic. (CO5)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)
- Q.23 Explain the concept of state space search in detail. How does it help in solving AI problems? Illustrate with an example. (CO1)
- Q.24 Discuss the various approaches to knowledge representation in AI. What are the advantages and limitations of each? (CO3)
- Q.25 What is the frame problem? Why is it important in AI? How can it be addressed in knowledge representation systems? (CO3)

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Time : 3 Hrs.

M.M. : 60

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (6x1=6)
- Q.1 Which of the following is not an AI technique? (CO1)
- a) Heuristic Search
 - b) Neural Networks
 - c) Bubble Sort
 - d) Natural Language Processing
- Q.2 In a state space search, what does each node represent? (CO1)
- a) A function b) A rule
 - c) A solution d) A state
- Q.3 Which search technique uses a heuristic to estimate the cost to reach the goal? (CO1)
- a) Depth-First Search b) Breadth-First Search
 - c) Best-First Search d) Linear Search

- Q.4 Which of the following is a characteristic of Hill Climbing? It _____. (CO2)
- Evaluates all possible paths
 - Moves in the direction of increasing value
 - Always finds global maximum
 - Stores all visited nodes
- Q.5 Which of these is a key challenge in knowledge representation? (CO3)
- High storage cost
 - Frame problem
 - Compilation error
 - Memory leak
- Q.6 In predicate logic, the statement "Sandeep is a teacher" is best represented as : (CO4)
- Sandeep = Teacher
 - Teacher(Sandeep)
 - Sandeep → Teacher
 - Teacher := Sandeep

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define "State Space Search" in Artificial Intelligence. (CO1)
- Q.8 Define the term "Heuristic Search". (CO1)
- Q.9 Semantic Networks are a method of representing knowledge in AI. (True / False) (CO3)

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- Q.10 In predicate logic, the Isa relationship is used to denote class membership. (True/False) (CO4)
- Q.11 Define "Control Knowledge" in rule-based systems. (CO5)
- Q.12 In _____ reasoning, the system starts with a goal and works backward to find supporting facts. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Discuss the main components of a production system in AI? (CO1)
- Q.14 Describe the working principle of the Hill Climbing algorithm. (CO2)
- Q.15 Explain the main phases of the Expert System Life Cycle. (CO4)
- Q.16 Explain the difference between an AI problem and a conventional programming problem. (CO1)
- Q.17 Write a simple fact and its representation in predicate logic. (CO4)
- Q.18 Distinguish between forward and backward reasoning? (CO5)
- Q.19 What are the key characteristics of expert systems? (CO4)

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